UNIT-4

LIBRARIES AND DATASETS

4.1. Jupyter Installation and Use

- Jupyter Notebook is an open-source web application that allows you to create and share documents that contain live code, equations, visualizations, and narrative text. Uses include data cleaning and transformation, numerical simulation, statistical modeling, data visualization, machine learning, and more.
- Jupyter supports over 40 different programming languages and Python is one of them. Python is required (Python 3.3 or greater, or Python 2.7) for installing the Jupyter Notebook.

Jupyter Notebook can be installed by using either of the two ways described below:

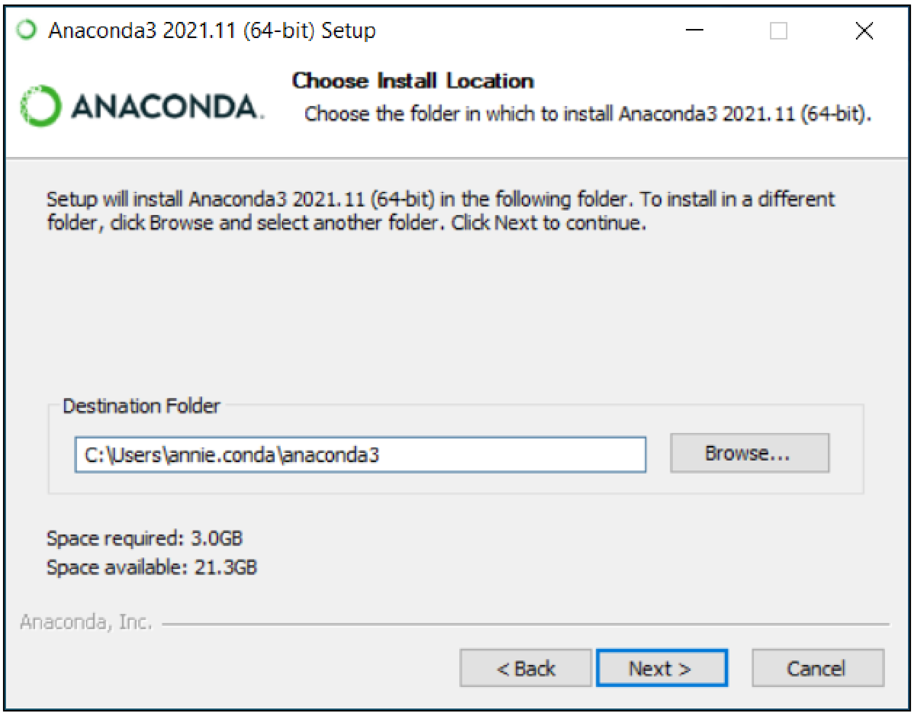
- **Using Anaconda:**

  Install Python and Jupyter using the Anaconda Distribution, which includes Python, the Jupyter Notebook, and other commonly used packages for scientific computing and data science. To install Anaconda, go through How to install Anaconda on Windows. And follow the instructions provided.

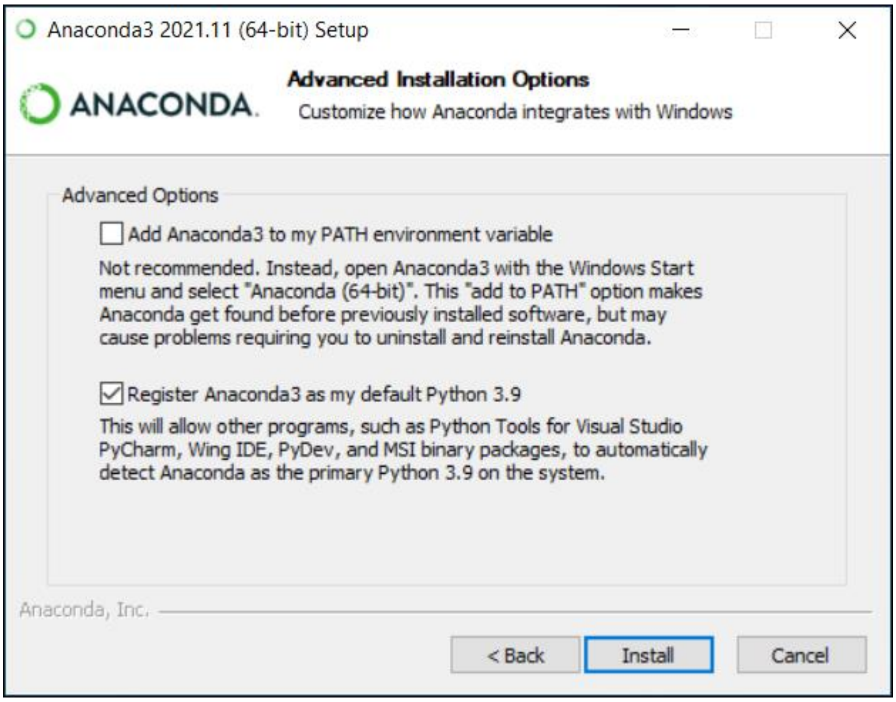
Installing Jupyter Notebook using Anaconda:

Anaconda is an open-source software that contains Jupyter, spyder, etc that is used for large data processing, data analytics, heavy scientific computing. Anaconda works for R and Python programming language. Spyder(sub-application of Anaconda) is used for Python. Opencv for python will work in spyder. Package versions are managed by the package management system called conda.

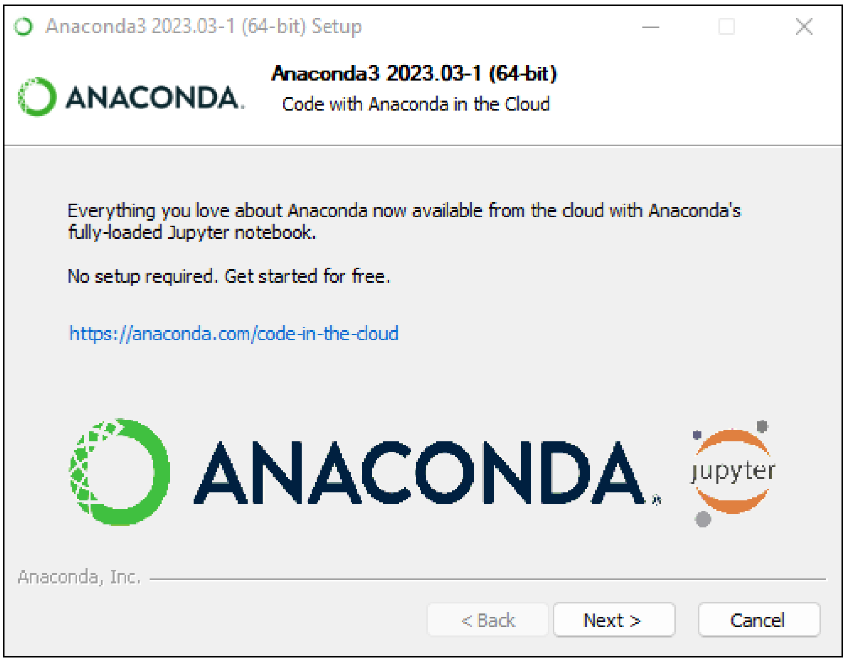
- So download Anaconda Visit: https://www.anaconda.com
- (Optional) Anaconda recommends verifying the integrity of the installer after downloading it.
- Go to your Downloads folder and double-click the installer to launch. To prevent permission errors, do not launch the installer from the Favorites folder.
- Click Next.
- Read the licensing terms and click I Agree.
- It is recommended that you install for Just Me, which will install Anaconda Distribution to just the current user account. Only select an install for All Users if you need to install for all users' accounts on the computer (which requires Windows Administrator privileges).
- Click Next.
- Select a destination folder to install Anaconda and click Next. Install Anaconda to a directory path that does not contain spaces or unicode characters. For more information on destination folders, see the FAQ.



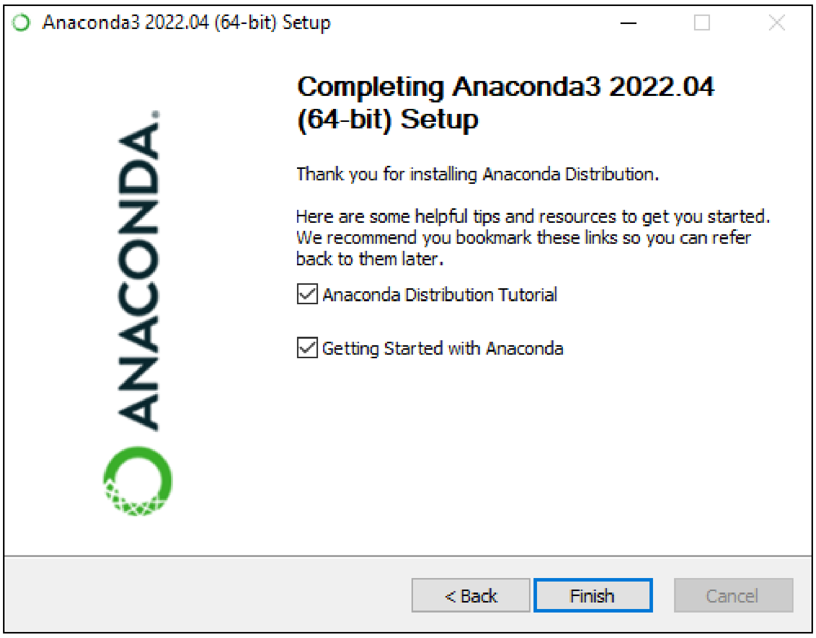
- Choose whether to add Anaconda to your PATH environment variable or register Anaconda as your default Python. We don’t recommend adding Anaconda to your PATH environment variable, since this can interfere with other software. Unless you plan on installing and running multiple versions of Anaconda or multiple versions of Python, accept the default and leave this box checked. Instead, use Anaconda software by opening Anaconda Navigator or the Anaconda Prompt from the Start Menu.



- Click Install. If you want to watch the packages Anaconda is installing, click Show Details.
- Click Next.
- Optional: To learn more about Anaconda's cloud notebook service, go to **https://www.anaconda.com/code-in-the-cloud**. Or click Continue to proceed.

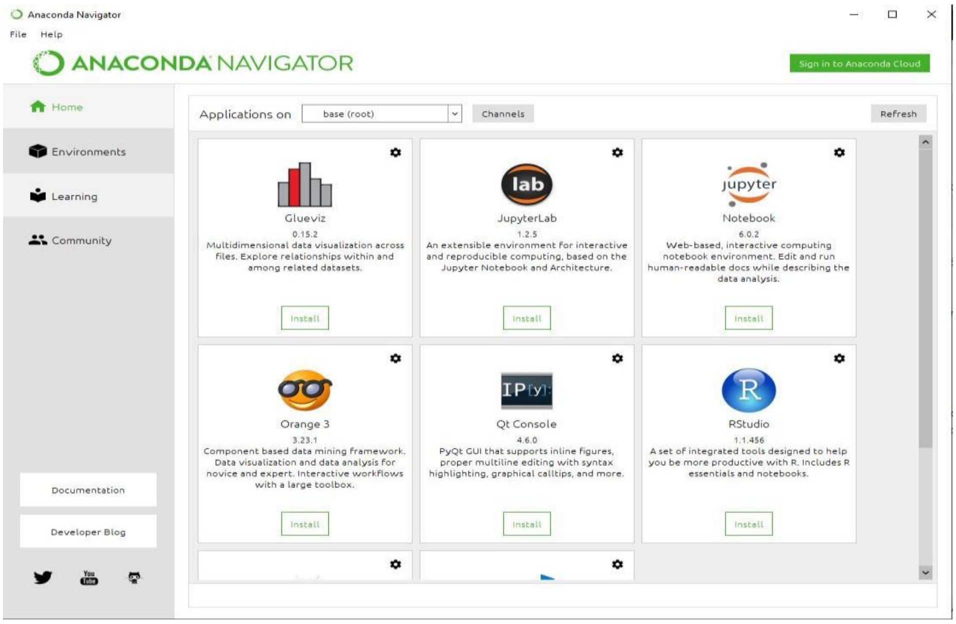


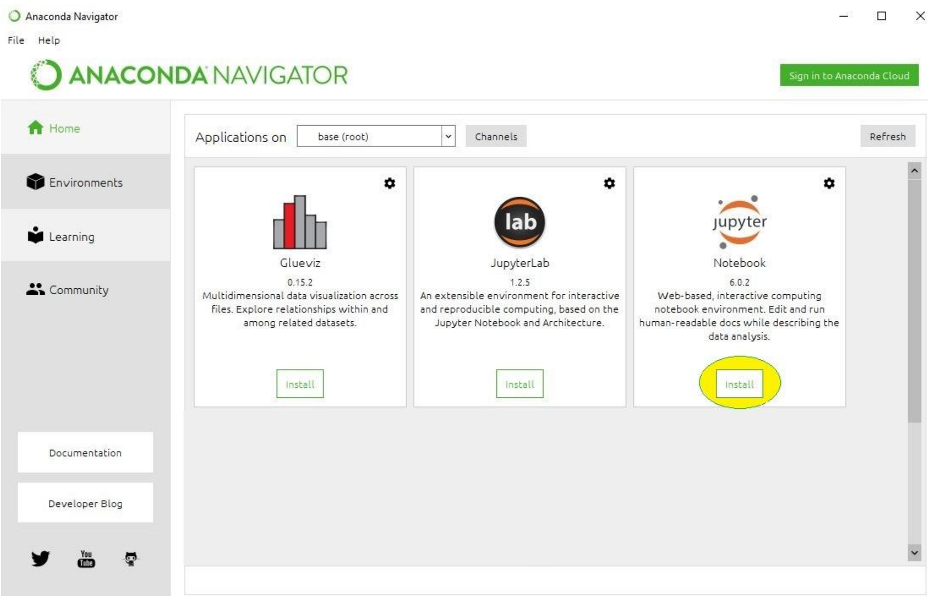
- After a successful installation you will see the "Thanks for installing Anaconda" dialog box:

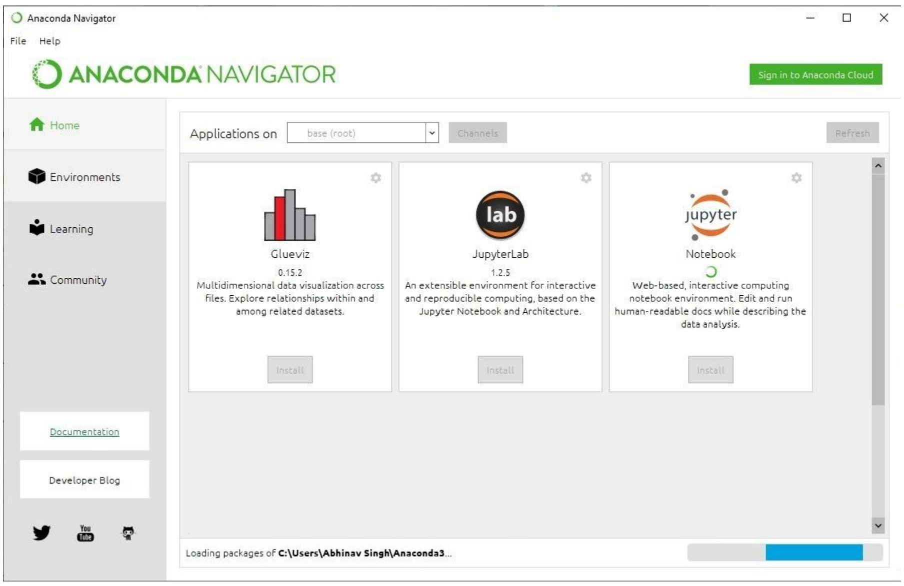


- If you wish to read more about Anaconda.org and how to get started with Anaconda, check the boxes "Anaconda Distribution Tutorial" and "Learn more about Anaconda". Click the Finish button.
- Verify your installation.
- **Launch Anaconda Navigator:**

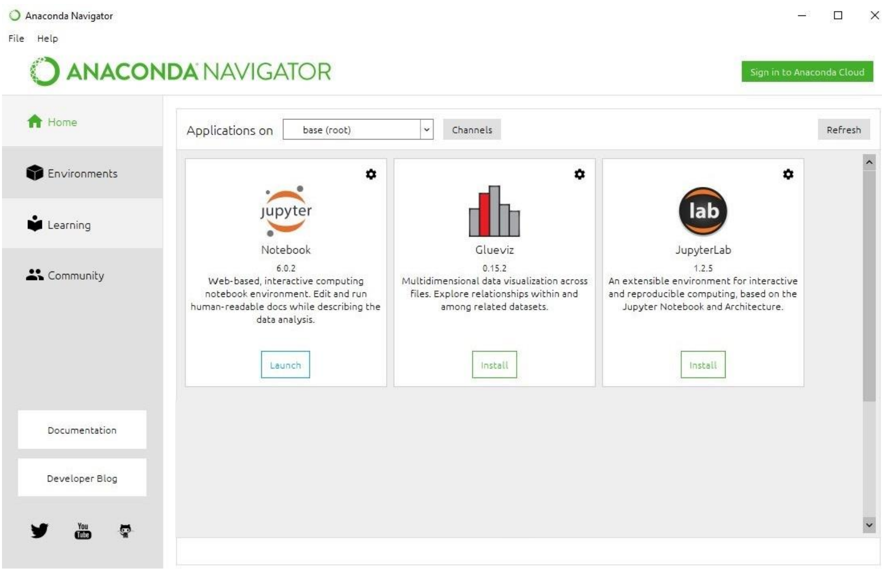
- **Click on the Install Jupyter Notebook Button:**







- **Loading Packages -> Finished Installation -> Launching Jupyter**



- **Installing Jupyter Notebook using pip:**

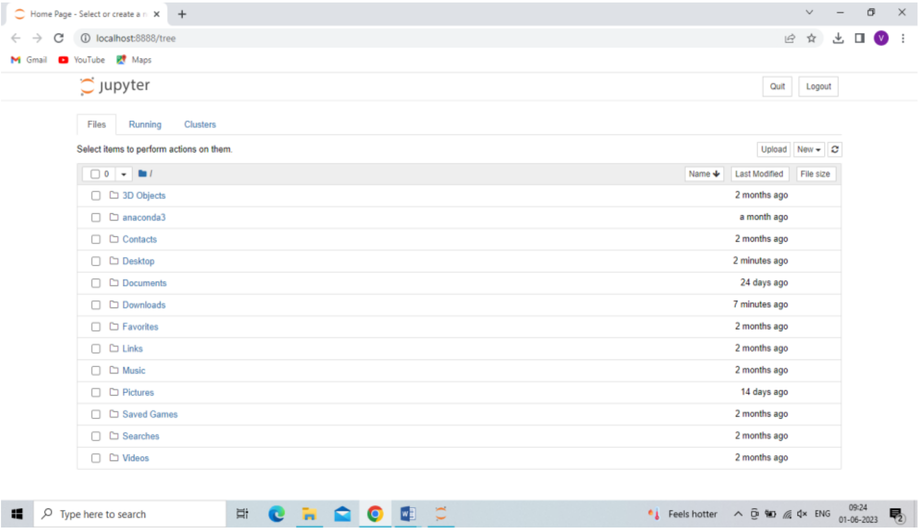
  PIP is a package management system used to install and manage software packages/libraries written in Python. These files are stored in a large "on-line repository" termed as Python Package Index (PyPI).
  pip uses PyPI as the default source for packages and their dependencies.
  To install Jupyter using pip, we need to first check if pip is updated in our system. Use the following command to update pip:
  python -m pip install --upgrade pip

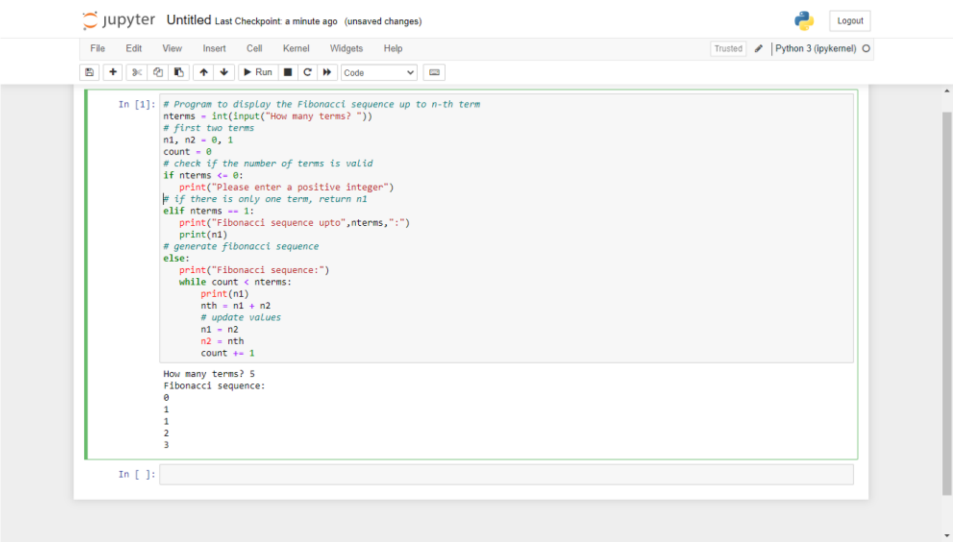
- **Launching Jupyter:**
  Use the following command to launch Jupyter using command line:
  jupyter notebook

Jupyter Uses:

- Programming Practice.
- Collaborating Across Projects and Tools.
- Data Organization and Cleaning.
- Data Visualization and Sharing.

- **Run code in Jupyter**





4.2. Datasets: Kaggle

- A dataset is a collection of structured or unstructured data that is organized and stored for analysis, processing, and information retrieval. It can be thought of as a set of observations or records, each representing a distinct entity or item.
- Datasets can come in various formats, including spreadsheets, databases, text files, images, videos, or any other form of digital information. They can be created for specific purposes, such as scientific research, machine learning, data analysis, or business intelligence.
- In the context of machine learning, datasets play a crucial role. They are used to train, validate, and test machine learning models. A typical machine learning dataset consists of input features or variables and their corresponding target or output values. By analyzing and learning from the patterns and relationships within the dataset, machine learning models can make predictions or perform tasks based on new, unseen data.

- Datasets can be obtained from various sources, including public repositories, research institutions, government agencies, private companies, or by collecting data directly through surveys, experiments, or sensors. It is important to ensure that datasets are representative, reliable, and appropriately processed to yield meaningful and accurate results in any data analysis or machine learning project.
- Inside Kaggle you'll find all the code & data you need to do your data science work. Use over 50,000 public datasets and 400,000 public notebooks to conquer any analysis in no time.

4.3. Python Libraries:

Python has a rich ecosystem of libraries and packages that provide additional functionality and extend the capabilities of the language. Here are some popular Python libraries across different domains:

```
import numpy as np
```

- **NumPy** (Numerical Python) is a fundamental library for scientific computing in Python. It provides support for large, multi-dimensional arrays and matrices, along with a collection of mathematical functions to operate on these arrays efficiently. Here are some key features and concepts related to NumPy
  - **ndarray:** NumPy's main object is the ndarray (n-dimensional array). It represents a grid of values, all of the same data type, indexed by a tuple of non-negative integers. The dimensions of an array are called axes, and the number of axes is known as the array's rank.
  - **Array Creation**: NumPy provides various functions to create arrays. For example, you can create an array from a Python list using the numpy.array() function. Other functions like numpy.zeros(), numpy.ones(), and numpy.arange() can be used to create arrays initialized with zeros, ones, or a range of values.

```
>>> x = np.array([[1, 2, 3], [4, 5, 6]], np.int32)
>>> type(x)
<class 'numpy.ndarray'>
>>> x.shape
(2, 3)
>>> x.dtype
dtype('int32')
```

- **Array Operations**: NumPy enables efficient element-wise operations on arrays, such as arithmetic operations (addition, subtraction, multiplication, division), exponentiation, and trigonometric functions. These operations are vectorized, meaning they are applied to the entire array rather than individual elements, resulting in improved performance.

```python
# method to insert element


def insert(arr, element):
  arr.append(element)


# Driver's code
if __name__ == '__main__':
  # declaring array and key to insert
  arr = [12, 16, 20, 40, 50, 70]
  key = 26

  # array before inserting an element
  print("Before Inserting: ")
  print(arr)

  # array after Inserting element
  insert(arr, key)
  print("After Inserting: ")
  print(arr)

  # Thanks to Aditi Sharma for contributing
  # this code
```

- **Universal Functions (ufuncs):** NumPy provides a large collection of universal functions (ufuncs) that operate element-wise on arrays, including mathematical functions (e.g., numpy.sin(), numpy.cos()), statistical functions (e.g., numpy.mean(), numpy.std()), logical operations (e.g., numpy.logical_and(), numpy.logical_or()), and more.

```python
# prompt: in  numpy Universal Functions (ufuncs)

import numpy as np
# ufuncs are element wise operations
# use the ufunc add to add two arrays
a = np.array([5, 7, 9, 8, 6, 4, 5])
b = np.array([6, 3, 4, 8, 9, 7, 1])
c = np.add(a, b)
print(c)
```

- **Indexing and Slicing:** NumPy allows indexing and slicing of arrays to access specific elements, rows, columns, or subarrays. This enables efficient data extraction and manipulation. Indexing starts at 0, and negative indices can be used to access elements from the end of the array.

```python
# prompt: Indexing and Slicing:

# Create a list
list1 = [10, 20, 30, 40, 50]

# Print the list
print(list1)

# Access the element at index 3
print(list1[3])

# Access the elements from index 1 to 3
print(list1[1:3])

# Access the elements from the beginning to index 3 (excluding 3)
print(list1[:3])

# Access the elements from index 1 to the end
print(list1[1:])

# Access the elements from index -4 to the end
print(list1[-4:])

# Access the elements from the beginning to index -4 (excluding -4)
print(list1[:-4])

# Access the elements at even indices
print(list1[::2])

# Access the elements at odd indices
print(list1[1::2])

# Access the elements in reverse order
print(list1[::-1])
```

- **Shape Manipulation:** NumPy provides functions to change the shape or dimensions of arrays. Functions like numpy.reshape(), numpy.flatten(), and numpy.transpose() allow you to manipulate the shape and layout of arrays according to your needs.

```python
# prompt: Shape Manipulation

import numpy as np

# Create a 3x3 array
X = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])

# Reshape the array to a 9x1 array
Y = X.reshape(9, 1)

# Print the reshaped array
print(Y)
```

- **Broadcasting:** Broadcasting is a powerful feature of NumPy that allows operations between arrays of different shapes, as long as they are compatible. Broadcasting enables implicit element-wise operations between arrays with different sizes, reducing the need for explicit loops and improving code readability.

```
# prompt: Broadcasting

import numpy as np

a = np.array([1, 2, 3])
b = np.array([10, 20, 30])
c = np.array([[100, 200, 300],
              [400, 500, 600]])

# Broadcasting a 1-D array with a 2-D array
print("Broadcasting a 1-D array with a 2-D array:")
print(a + c)

# Broadcasting a 2-D array with a 3-D array
print("\nBroadcasting a 2-D array with a 3-D array:")
print(b + c)
```

- **Linear Algebra:** NumPy provides a rich set of functions for linear algebra operations, such as matrix multiplication (numpy.matmul() or @ operator), matrix inversion (numpy.linalg.inv()), eigenvalue computation (numpy.linalg.eig()), and more.

```
# prompt: Linear Algebra

import numpy as np

# Create a 3x3 matrix
A = np.array([[1, 2, 3], [4, 5, 6], [7, 8, 9]])

# Calculate the determinant of the matrix
det_A = np.linalg.det(A)

# Print the determinant
print("The determinant of matrix A is:", det_A)
```

- **Random Number Generation:** NumPy includes functions to generate random numbers from various probability distributions. The numpy.random module provides functions like numpy.random.rand(), numpy.random.randn(), and numpy.random.randint() for generating random arrays or numbers.

```
# prompt: Random Number Generation

import random
random_number = random.random()
print(random_number)
```

- **Performance Optimization:** NumPy's underlying implementation is in C, which makes it highly optimized for performance. It provides efficient and fast numerical operations, making it a preferred choice for scientific computing and numerical computations.
- NumPy is widely used in various domains, including scientific research, data analysis, machine learning, and computational mathematics. It serves as a foundational library for many other libraries and packages in the Python ecosystem, enabling efficient data manipulation, numerical computations, and advanced mathematical operations.

- **Pandas:** pandas is a powerful library for data manipulation and analysis. It offers data structures such as dataframes and Series, which enable easy handling, cleaning, and exploration of structured data. Here are some key features and concepts related to Pandas:

```python
import pandas as pd

# Create a DataFrame
data = {'Name': ['John', 'Mary', 'Bob'], 'Age': [20, 25, 30]}
df = pd.DataFrame(data)

# Print the DataFrame
print(df)
```

- **DataFrame:** The central data structure in Pandas is the DataFrame. It represents a two-dimensional table of data with labeled rows and columns. Each column in a DataFrame is called a Series, which is a one-dimensional labeled array. DataFrames can contain heterogeneous data types (e.g., numbers, strings, dates) and handle missing values.
- **Data Input and Output:** Pandas provides functions to read data from various file formats, such as CSV, Excel, SQL databases, JSON, and more. The pandas.read_csv() function, for

```python
import pandas as pd

# Read a CSV file
df = pd.read_csv('data.csv')

# Read a JSON file
df = pd.read_json('data.json')

# Read an Excel file
df = pd.read_excel('data.xlsx')

# Read a text file
df = pd.read_table('data.txt')

# Write a DataFrame to a CSV file
df.to_csv('output.csv')

# Write a DataFrame to a JSON file
df.to_json('output.json')

# Write a DataFrame to an Excel file
df.to_excel('output.xlsx')

# Write a DataFrame to a text file
df.to_string('output.txt')
```

example, is commonly used to load data from a CSV file into a DataFrame. Similarly, Pandas offers functions like to_csv(), to_excel(), and to_sql() to export data from a DataFrame to different formats.

```
import pandas as pd

# Create a DataFrame
df = pd.DataFrame({'Name': ['John', 'Mary', 'Peter'], 'Age': [20, 25, 30]})

# Select rows where Age is greater than 25
df_filtered = df[df['Age'] > 25]

# Select specific columns
selected_columns = df[['Name', 'Age']]

# Update a specific cell
df.loc[0, 'Name'] = 'Jane'

# Delete a specific row
df.drop(0, inplace=True)

# Print the DataFrame
print(df)
```

- **Data Selection and Manipulation:** Pandas offers powerful indexing and selection capabilities to extract, filter, and manipulate data in a DataFrame. You can use column names or labels, as well as numerical indexing, to access specific data. Functions like loc[] and iloc[] allow for label-based and integer-based indexing, respectively. Operations like filtering rows based on conditions, selecting specific columns, or applying functions across the data are also straightforward with Pandas.
- **Data Cleaning and Preprocessing:** Pandas provides a variety of functions to handle missing data, duplicate values, and outliers. Functions like dropna(), fillna(), and duplicated() assist in

```
import pandas as pd
import numpy as np

# Create a sample DataFrame with missing values
df = pd.DataFrame({
    "column1": [1, 2, None, 4, 5],
    "column2": ["a", None, "c", "d", None],
    "column3": [True, False, True, None, True],
})

# Drop rows with missing values
df.dropna(inplace=True)

# Fill missing values with a specific value
df.fillna(value=0, inplace=True)

# Fill missing values with the mean of each column
df.fillna(value=df.mean(), inplace=True)

# Drop columns with missing values
df.dropna(axis=1, inplace=True)
```

removing or imputing missing values, identifying duplicates, and handling outliers. Additionally, Pandas offers functions for data transformation, such as sorting, merging, reshaping, and aggregating data, to prepare it for analysis.

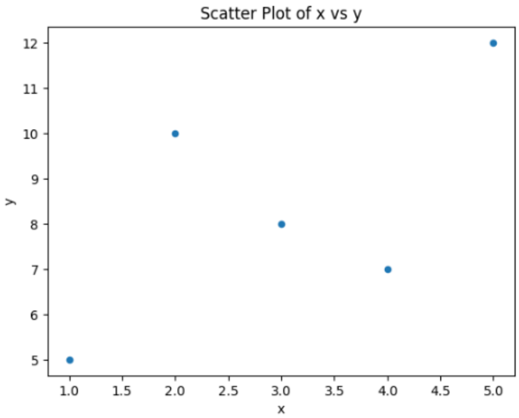
- **Descriptive Statistics:** Pandas enables the calculation of various descriptive statistics on numerical data in a DataFrame. Functions like mean(), median(), std(), min(), and max() provide summary statistics for columns or across the entire DataFrame. The describe() function generates a comprehensive summary of the distribution of each column, including count, mean, standard deviation, minimum, quartiles, and maximum values.

```python
import pandas as pd

# Create a dataframe
df = pd.DataFrame({
    "column1": [1, 2, 3, 4, 5],
    "column2": [6, 7, 8, 9, 10]
})

# Print descriptive statistics
print(df.describe())
```

```
        column1    column2
count  5.000000   5.000000
mean   3.000000   8.000000
std    1.581139   1.581139
min    1.000000   6.000000
25%    2.000000   7.000000
50%    3.000000   8.000000
75%    4.000000   9.000000
max    5.000000  10.000000
```

- **Data Visualization:** Pandas integrates well with other Python visualization libraries, such as Matplotlib and Seaborn, allowing for easy data visualization. Pandas provides built-in plotting

```python
import pandas as pd
import matplotlib.pyplot as plt

# Create a dataframe
df = pd.DataFrame({'x': [1, 2, 3, 4, 5], 'y': [5, 10, 8, 7, 12]})

# Plot the data using a scatter plot
df.plot.scatter(x='x', y='y')

# Add a title and labels to the plot
plt.title('Scatter Plot of x vs y')
plt.xlabel('x')
plt.ylabel('y')

# Show the plot
plt.show()
```

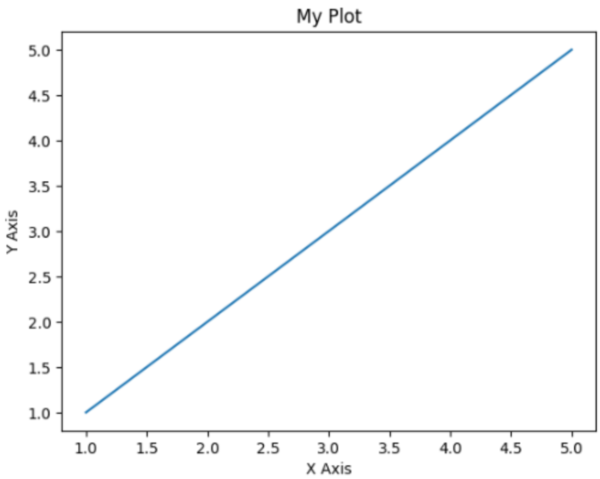


functions, including line plots, bar plots, histograms, scatter plots, and more. These functions simplify the process of creating basic visualizations directly from a DataFrame or Series.

- **Time Series Functionality:** Pandas has extensive support for time series data, making it suitable for analyzing and manipulating temporal data. It provides specialized data structures, such as the DatetimeIndex and PeriodIndex, along with functions to resample, interpolate, shift, and aggregate time series data. Pandas also offers date range generation and supports time zone handling.
- **Grouping and Aggregation:** Pandas allows for grouping data based on one or more variables and performing aggregations on those groups. The groupby() function is used to create groups, and then aggregate functions like sum(), mean(), count(), and apply() can be applied to calculate statistics on each group. This functionality is particularly useful for analyzing data by categories or performing group-level operations.
- **Integration with NumPy and Scikit-Learn:** Pandas seamlessly integrates with other libraries, such as NumPy and scikit-learn. It can convert between Pandas data structures and NumPy arrays efficiently. This integration enables smooth data transformations and preprocessing before feeding the data into machine learning models built with scikit-learn.

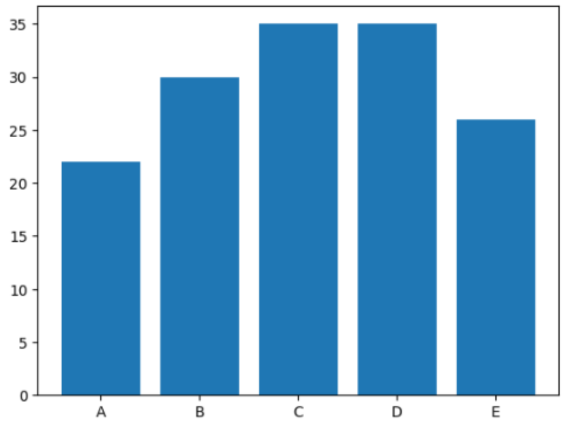
- **Matplotlib:** Matplotlib is a plotting library that provides a wide range of options for creating static, animated, and interactive visualizations. It is widely used for generating charts, histograms, scatter plots, and other types of plots.
  - **Pyplot Interface:** Matplotlib's pyplot module provides a MATLAB-like interface for creating and customizing plots. It enables users to quickly generate basic plots by using functions like plot(), scatter(), bar(), hist(), and imshow(). Pyplot functions allow for easy customization of plot elements such as labels, titles, legends, colors, and line styles.
  - **Figure and Axes:** A Figure is a top-level container that holds all the elements of a plot, including one or more Axes objects. Axes represent an individual plot with a specific set of coordinates. Multiple Axes can be arranged within a single Figure, allowing for the creation of subplots or complex layouts. The Figure and Axes structure provides fine-grained control over plot composition and appearance.
  - **Line Plots:** Matplotlib offers various types of line plots, including line graphs, step plots, and scatter plots. Line plots are commonly used for visualizing trends, time series data, and continuous variables. Users can specify line styles, markers, colors, and annotations to enhance the visual representation of the data.

```
import seaborn as sns
sns.lineplot(x=[1,2,3,4,5], y=[2,4,5,8,10])
```



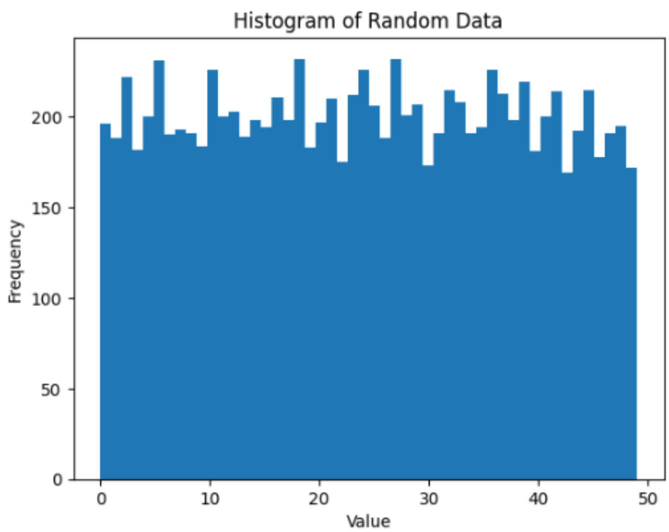
- **Bar Plots:** Bar plots in Matplotlib are useful for comparing different categories or groups. They can represent both categorical and numerical data. Matplotlib supports vertical and horizontal bar plots, grouped and stacked bar plots, and customization of bar widths, colors, and labels.

```
# prompt: Bar Plots

import matplotlib.pyplot as plt
import numpy as np

x = np.array(["A", "B", "C", "D", "E"])
y = np.array([22, 30, 35, 35, 26])

plt.bar(x, y)
plt.show()
```



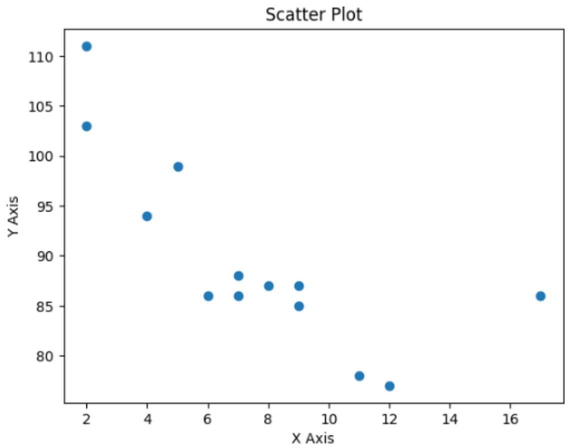
- **Histograms:** Histograms are effective for visualizing the distribution of a continuous variable. Matplotlib allows users to create histograms with customizable bin sizes, bin edges, and

```
# prompt: Histograms

import matplotlib.pyplot as plt
import numpy as np

# Create data
x = np.random.randint(0, 50, 10000)

# Create a histogram
plt.hist(x, bins=50)

# Add labels and title
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.title("Histogram of Random Data")

# Show the plot
plt.show()
```



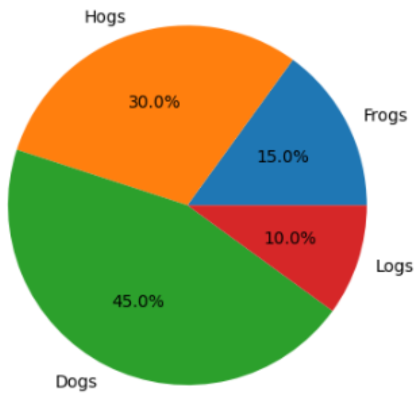
normalization options. Histograms can provide insights into the underlying data distribution, identify outliers, and highlight key statistical properties.

```python
import matplotlib.pyplot as plt

# Create data
x = [5, 7, 8, 7, 2, 17, 2, 9, 4, 11, 12, 9, 6]
y = [99, 86, 87, 88, 111, 86, 103, 87, 94, 78, 77, 85, 86]

# Create the scatter plot
plt.scatter(x, y)

# Add labels and title
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.title("Scatter Plot")

# Show the plot
plt.show()
```



- **Scatter Plots:** Scatter plots are ideal for visualizing relationships between two continuous variables. Matplotlib provides functions to create scatter plots with options for marker styles, sizes, colors, and transparency. Scatter plots are useful for identifying patterns, clusters, or correlations in data.

```python
import matplotlib.pyplot as plt

# Data to plot
labels = 'Frogs', 'Hogs', 'Dogs', 'Logs'
sizes = [15, 30, 45, 10]

# Create a pie chart
plt.pie(sizes, labels=labels, autopct='%1.1f%%')

# Show the pie chart
plt.show()
```



- **Pie Charts:** Matplotlib supports the creation of pie charts to display proportions or percentages of categorical data. Pie charts can be customized with colors, labels, and explode options to emphasize specific slices. Matplotlib also allows for exploded pie charts and donut charts.
- **Annotations and Labels:** Matplotlib enables users to add annotations, text, and labels to plots. Annotations can be used to highlight specific data points, provide additional context, or add descriptions. Matplotlib supports custom text positioning, font styles, arrow annotations, and LaTeX rendering for mathematical expressions.
- **Customization and Styling:** Matplotlib provides extensive customization options to fine-tune the appearance of plots. Users can control axis limits, ticks, labels, grid lines, legends, color maps, line styles, and plot backgrounds. Matplotlib supports the use of style sheets to apply predefined visual themes or create custom styles.

```
# Save the chart to a file.
chart.save('chart.png')

# Export the chart as a base64-encoded PNG image.
chart_b64 = chart.to_base64()
```

- **Saving and Exporting:** Matplotlib allows users to save plots in various formats, including PNG, JPEG, PDF, SVG, and more. Plots can be saved programmatically or interactively using the GUI provided by Matplotlib. This feature facilitates the integration of Matplotlib-generated plots into reports, presentations, or web applications.

- **Scikit-learn:** scikit-learn is a popular machine-learning library that provides various algorithms and tools for classification, regression, clustering, dimensionality reduction, and model selection. It also offers utilities for data preprocessing, model evaluation, and cross-validation.
- **Tensorflow:** TensorFlow is an open-source library primarily used for deep learning and neural network-based computations. It provides a flexible platform for building and training machine learning models, especially those involving large-scale datasets and complex architectures.
- **Keras:** Keras is a high-level deep-learning library that runs on top of tensorflow. It offers a user-friendly interface for defining and training neural networks, making it easier to experiment and iterate on different models.
- **Pytorch:** pytorch is another popular deep-learning library that emphasizes flexibility and dynamic computation graphs. It provides efficient tensor operations and automatic differentiation, making it suitable for research and development in the field of deep learning.
- **Opencv:** OpenCV (Open Source Computer Vision Library) is a powerful library for computer vision and image processing tasks. It offers a wide range of functions and algorithms for image and video analysis, object detection, feature extraction, and more.
- **NLTK:** NLTK (Natural Language Toolkit) is a library for natural language processing. It provides tools and resources for tasks such as tokenization, stemming, tagging, parsing, and sentiment analysis, making it valuable for text-based applications.
- **Django:** Django is a high-level web framework that simplifies the process of building web applications in Python. It follows the Model-View-Controller (MVC) architectural pattern and provides features like URL routing, template rendering, database ORM (Object-Relational Mapping), and user authentication.

These are just a few examples of the numerous Python libraries available. Depending on your specific needs and interests, many other libraries and packages can cater to different domains, such as image processing, natural language processing, data visualization, network programming, and more.